

Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical AI

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Abstract—Hospital FL faces gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under multi-year retention. ZKFL-PQ’s contribution is a *defense-in-depth composition*: NIST ML-KEM-768 transport; an Unruh NIZK (default $r=128$) of ℓ_2 -bounded updates in language $\mathcal{L}_\tau^{\text{bind}}$ with an Enc-consistency gadget for BFV coins; full-vector BFV aggregation opened by (t, n) -threshold *partial* decryption (sk never reconstructed); and a default post-ZKP coordinate-wise median that closes sign-flip/backdoor residuals that pure ℓ_2 gates must accept. A public artifact ships fused SEAL+threshold HE, Lean/Python/EasyCrypt combinatorial Unruh checks, and WiMob-oriented payload knobs. Empirically: multi-seed synthetic baselines, scale $N=20/T=30$, UCI Breast Cancer, full-resolution PneumoniaMNIST, and a backdoor ASR win for ZKP+median vs. FedAvg/ ℓ_2 -only. We complement—not replace—Beskar-class secure aggregation.

Index Terms—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness

I. INTRODUCTION

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4].

a) Contributions.

- 1) **Composition punchline**: bind Unruh norm proofs to BFV ciphertexts via $\mathcal{L}_\tau^{\text{bind}}$ + Enc-consistency, open with threshold partial decrypt, then robust-aggregate (default median)—each layer covers a distinct threat (Table I).
- 2) Concrete Unruh ($r=128$), BFV presets, and WiMob payload knobs (r , HE degree) with a named soundness/privacy theorem supported by Lean/Python/QROM.ec combinatorial checks.

TABLE I

THREAT COVERAGE BY LAYER (COMPOSITION IS THE CONTRIBUTION).

Threat	ML-KEM	ZKP ℓ_2	Enc-cons.	Thr. open	Median
HNDL / PQ transit	yes	—	—	—	—
Oversized update	—	yes	—	—	—
Proof/ct split	—	bind	yes	—	—
Curious aggregator	—	—	—	yes	—
Sign-flip / backdoor	—	no [‡]	—	—	yes

[‡]In-bound by design; Table IV reports 0% detection.

- 3) Reproducible artifact [12]: full-vector encrypt+prove, crypto-only accept, fused SEAL+threshold path, no monolithic server sk .
- 4) Empirics that make the niche sharp: large-norm vs. sign-flip honesty; backdoor ASR cut by ZKP+median (Fig. 3); UCI + full-res MedMNIST; scale $N=20$, $T=30$.

II. THREAT MODEL AND CLAIMS

Adversary: (i) network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f < N/2$ Byzantine clients; (iii) curious aggregator that sees protocol messages but does not hold sk alone; (iv) up to $t-1$ colluding threshold decryptors.

Claims: ML-KEM-768 transport (FIPS 203 Level 3) [5]; norm soundness for oversized witnesses under SIS + (Unruh)/RO assumptions; ciphertext binding so a proof cannot be replayed on a different ct; aggregator privacy via Shamir-shared sk and partial decryption $\mu_i = c_1 \star (\lambda_i \cdot s_i)$.

Scope. We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. ZKFL-PQ is the complementary *norm gate + multi-hospital decrypt + robust post-filter* stack; pure ℓ_2 alone does not stop sign-flip or in-bound backdoors (Table I).

III. RELATED WORK

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP [8]; we position ZKFL-PQ as a complementary norm-gate and threshold-decrypt path, not a replacement. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Multi-Krum remains

TABLE II
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	–	secure agg. stack
Patel [9]	lattice HE	–	HE FL
Lu et al. [10]	hybrid PQ	–	HE FL
Krum [3]	–	statistical	–
zkML [11]	–	inference ZK	–
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

strong against low-norm adversaries—hence our default post-ZKP median/Krum composition rather than claiming ℓ_2 alone.

IV. PROTOCOL DESIGN

A. Round Structure

Each round t : the server broadcasts $w^{(t)}$. Client i runs local SGD on D_i , quantizes to $\tilde{w}_i$, encrypts under BFV public key pk , proves $\|\tilde{w}_i\|_2 \leq \tau$ bound to $\mathbf{ct}_i$ (and Enc-consistency of coins), and sends an ML-KEM-wrapped payload. The server verifies proofs, threshold-opens each accepted ciphertext, and applies coordinate-wise median (default) before the model step (Algorithm 1).

B. Quantization

Quantize clips coordinates to $[-c, c]$, scales by s , and rounds into $\mathbb{Z}_t$ (artifact defaults $c=1$, $s=10^3$). The public threshold τ applies after quantization; adaptive rules $\tau^{(t)} = \hat{\mu} + k\hat{\sigma}$ are compatible. Reported runs use fixed τ (synthetic 5; medical 8–12).

C. Bound Norm Language

Definition 1 (Bound language). Let $C = \text{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w} \parallel \mathbf{r}) \bmod q'$ and $\mathbf{ct} = \text{Enc}_{pk}(\tilde{w}; \rho)$. Then

$$\mathcal{L}_\tau^{\text{bind}} = \{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \text{Commit}(\tilde{w}; \mathbf{r}), \mathbf{ct} = \text{Enc}(\tilde{w}; \rho), \|\tilde{w}\|_2 \leq \tau\}. \quad (1)$$

a) Σ -protocol [6].: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C \parallel \mathbf{ct} \parallel T \parallel \tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z} \parallel \mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \leq B$. Rejection sampling yields HVZK in the classical ROM.

b) *Unruh-oriented NIZK*.: Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default $r=128$; latency-sensitive demos often 32–64) with invertible-RO records. Combinatorial uniqueness / 2^{-r} lemmas are machine-checked in Lean 4; executable game hops and an in-repository EasyCrypt QROM interface (formal/easycrypt/QROM.ec) accompany the artifact.

c) *Enc-consistency*.: Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ -gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk, closing split attacks that pair a benign proof with a mismatched ciphertext.

TABLE III
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768	$n=256, k=3, q=3329$	FIPS 203 L3 [5]
BFV degree	$n=512 / 4096$	Demo vs. <code>classic128</code> (Classic-128 class)
BFV moduli	$q \approx 2^{40}, t=2^{12}$	NumPy encoder; optional estimator / SEAL
Threshold Unruh reps	$(t, n)=(2, 3)$ partial decrypt $r=128$ default	Server $sk=\perp$ Demos $r=64$ for latency; see Table V
Post-ZKP agg	median (default)	Multi-Krum / mean overrides
Packing	full-vector chunks	All d coordinates encrypted

Algorithm 1 ZKFL-PQ round (default composition)

Require: $w^{(t)}$, τ , KEM (ek_S, dk_S) , HE pk , decryptor shares

- 1: Broadcast $w^{(t)}$
- 2: **for** each client i **do**
- 3: $\tilde{w} \leftarrow \text{Quantize}(\text{LocalSGD}(w^{(t)}, D_i))$
- 4: $\mathbf{ct}_i \leftarrow \text{Enc}_{pk}(\tilde{w}; \rho_i)$; prove Enc-consistency of ρ_i
- 5: $\pi_i \leftarrow \text{Prove}_{\text{Unruh}}((\mathbf{ct}_i, \tau), \tilde{w})$
- 6: send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i)$
- 7: **end for**
- 8: Keep i iff cryptographic verify returns 1
- 9: For each accepted i : threshold-open $\mathbf{ct}_i$ to recover Δ_i
- 10: $\bar{\Delta} \leftarrow \text{Median}(\{\Delta_i\})$ ▷ or Multi-Krum; not plaintext mean alone
- 11: $w^{(t+1)} \leftarrow w^{(t)} + \bar{\Delta}$

D. BFV Aggregation and Threshold Opening

Additive BFV yields $\text{Dec}(\sum_i \text{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. After key generation, secret coefficients are Shamir-shared and the monolithic sk is erased. Opening uses Lagrange-weighted partial decryptions with smudging: s is never reconstructed for production decrypt. An optional TenSEAL/SEAL sidecar runs in the same process as threshold BFV (`FusedSealThresholdHE`); primary open remains partial-decrypt, while SEAL provides a certified encoder path and consistency check.

Theorem 1 (Composition soundness and threshold privacy). Assume SIS binding of Commit, RLWE semantic security of BFV, and RO modeling of H . Then: (i) honest $\|\tilde{w}\|_2 \leq \tau$ proofs are accepted except for rejection-sampling aborts; (ii) an oversized witness is accepted with probability at most $\approx 2^{-r}$ (binary Unruh) plus SIS advantage; (iii) Enc-consistency plus digesting $\mathbf{ct}$ into challenges prevent attaching a valid proof to a mismatched ciphertext; (iv) the aggregator and any $t-1$ decryptors learn nothing beyond what t partial opens reveal. Combinatorial Unruh uniqueness/ 2^{-r} is machine-checked in Lean 4 with executable Python game hops and in-repo EasyCrypt QROM.ec (no claim of a full SHA3-QROM production proof).

TABLE IV
MEAN $\pm$ STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0 $\pm$ 1.0	41.0 $\pm$ 13.2	0%	0%
Clip@ τ	100 $\pm$ 0	40.3 $\pm$ 12.7	—	—
Multi-Krum	100 $\pm$ 0	100 $\pm$ 0	—	—
HE-only	22.1 $\pm$ 2.1	19.8 $\pm$ 9.5	0%	0%
ZKP-only	84.9 $\pm$ 30.2	40.9 $\pm$ 13.2	97%	0% [‡]
Hybrid+mean	87.0 $\pm$ 26.0	40.1 $\pm$ 12.6	97%	0% [‡]
Hybrid+median	100 $\pm$ 0	100 $\pm$ 0	97%	0% [‡]

[‡]Sign-flip respects τ — ℓ_2 proofs must accept (0% crypto detection). Hybrid+mean keeps plaintext mean after accept; Hybrid+median is the default composition and restores accuracy [12].

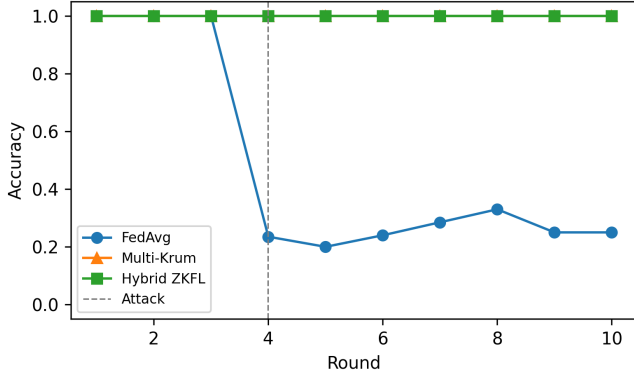


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

a) *Implementation.*: The public repository [12] realizes Algorithm 1: shared prove/encrypt dimension, ciphertext-bound challenges, crypto-only accept, full-vector packing, Enc-consistency, fused SEAL+threshold HE, and default median. Scripts pin seeds and emit JSON.

V. EVALUATION

A. Synthetic Multi-Baseline Study

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet $\alpha=0.5$), $N=5$, MLP with $d=108,996$, 10 rounds, seeds $\{42, \dots, 46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@ τ , Multi-Krum ($f=1$), HE-only, ZKP-only, Hybrid+mean (privacy path without robust agg), and Hybrid+median (default composition).

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table IV, Fig. 1–2). Sign-flip within τ defeats pure ℓ_2 and Hybrid+mean (0% detection by design, Table I), while Multi-Krum and default Hybrid+median recover accuracy by pairwise / coordinate-wise robust aggregation after cryptographic accept.

B. Scale, Medical, and Backdoor Studies

A compact-model scale study with $N=20$ clients and $T=30$ rounds reproduces the same qualitative pattern under both

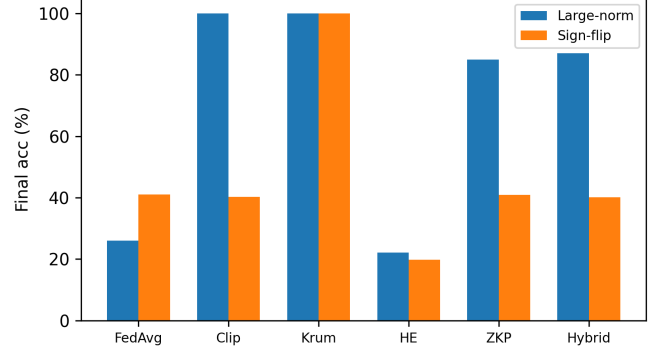


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

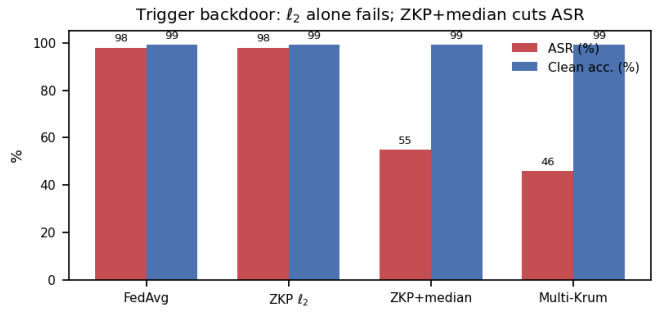


Fig. 3. Backdoor ASR vs. clean accuracy (default ZKP+median closes the ℓ_2 residual).

attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator. Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with $d=1554$), the target stack—Unruh with $r=64$ (library default 128), full-vector fused HE, (2, 3) partial threshold decrypt, and post-ZKP median—yields accuracies 88.6% $\rightarrow$ 93.0% $\rightarrow$ **93.9%** across three rounds while rejecting a scripted oversized client (Fig. 4). Mean round time is ≈ 4 –9 s and per-round payload ≈ 1.4 –1.7 MB on a laptop CPU. Full-resolution PneumoniaMNIST uses native 28×28 images (784 pixels, no learned or random projection), a compact head, and 25 HE chunks; accuracy reaches about 67.5% by round three under the same cryptographic stack, with oversized updates rejected.

Trigger backdoors whose poisoned updates remain inside τ are accepted by pure ℓ_2 ZKPs (ASR $\approx 98\%$, matching FedAvg). Default ZKP+median cuts ASR to $\approx 55\%$ while preserving clean accuracy $\approx 99\%$; Multi-Krum alone reaches $\approx 46\%$ ASR (Fig. 3). The win story is composition: cryptographic accept for oversized updates, statistical robust aggregation for in-bound poisons.

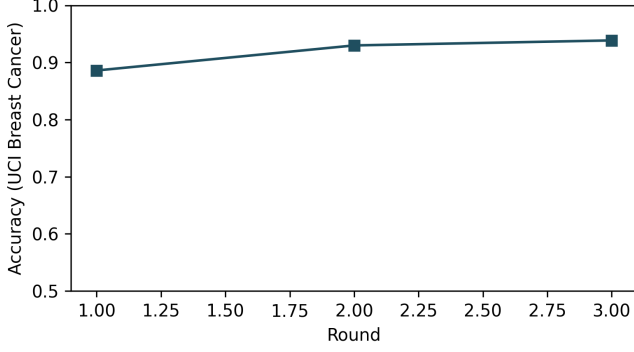


Fig. 4. UCI Breast Cancer accuracy under the full target stack.

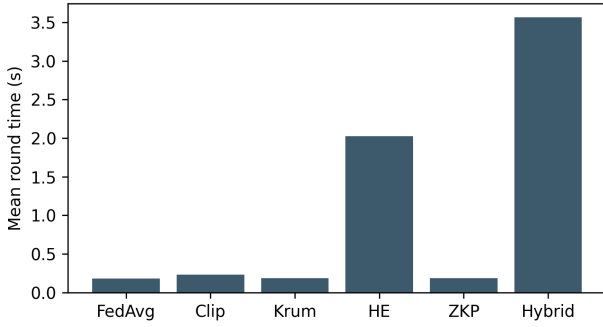


Fig. 5. Mean round time (synthetic hybrid $\sim 20\times$ FedAvg).

C. Communication and WiMob Knobs

Full-vector HE dominates bandwidth. Uplink scales as $O(N\lceil d/n \rceil n \log q)$ bits plus Unruh $O(r \cdot (d + \ell))$. Table V: $r=128$ roughly doubles the proof component vs. $r=64$; `classic128` ($n=4096$) targets the claimed Classic-128 HE class at about $2\times$ ciphertext volume for the UCI-sized model.

VI. SECURITY DISCUSSION

Honest clients with $\|\tilde{w}\|_2 \leq \tau$ produce Unruh sessions that pass algebraic and norm checks except for rejection-sampling aborts (bounded retries in the artifact). If $\|\tilde{w}\|_2 \gg \tau$, sessions with challenge bit 1 fail with overwhelming probability under the Gaussian mask design; soundness therefore depends on the fraction of 1-challenges across r repetitions. The library default $r=128$ targets a ≈ 128 -bit combinatorial Unruh class; demos may reduce r for CPU latency at linear proof-size cost. Binding H to ciphertext bytes, together with the Enc-consistency gadget for ρ , prevents a benign proof from being attached to a different ciphertext. BFV semantic security (RLWE) hides individual updates from parties lacking sk ; with Shamir (t, n) , the aggregator and any $t-1$ decryptors cannot open. We do not claim differential privacy; sites that need DP can compose mechanisms as in Beskar [8].

TABLE V
PAYLOAD / SECURITY KNOBS (SCALED ACCOUNTING).

Setting	Msg/round	Notes
Synthetic hybrid	~ 100 KB	Reference
Medical $r=64$, $n=512$	1.4–1.7 MB	Demo CPU
Medical $r=128$, $n=512$	≈ 2.0 –2.5 MB	Unruh default
Medical $r=128$, $n=4096$	≈ 3 –4 MB	Classic-128 class
Plaintext FL ($d \approx 10^5$)	~ 4.3 MB	No crypto

a) *Layering.*: Table I summarizes the division of labor: ML-KEM for PQ transit, ZKP+binding+Enc-consistency for oversized and split attacks, threshold open for curious aggregators, median for in-bound Byzantine. None of the layers alone meets the joint medical threat model; the contribution is their composition with explicit binding.

b) *Deployment.*: In a hospital consortium, each site runs a client; a neutral coordinator verifies proofs; t -of- n share holders (for example a DPO and two hospital CISOs) jointly open per-client ciphertexts; median aggregation yields the round update. Daily or weekly training absorbs the observed $\sim 20\times$ CPU overhead vs. plaintext FedAvg (Fig. 5). For imaging-scale d , HE chunks and Unruh proofs grow with d ; tabular/compact models (UCI) validate the crypto pipeline first.

VII. LIMITATIONS

Research-grade residuals remain: a full SHA3 instantiation inside a production EasyCrypt QROM library, and larger imaging CNNs without compact heads. The default artifact already delivers binding, full-vector HE, threshold privacy, fused SEAL+threshold, and post-ZKP median.

VIII. CONCLUSION

ZKFL-PQ’s punchline is defense-in-depth for medical FL: ML-KEM-768, ciphertext-bound Unruh+Enc-consistency, threshold BFV partial open, and default median—with honest measurement where ℓ_2 alone fails. Artifact: <https://github.com/edlansiaux/pq-zkfl-medical>.

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